

Shyam Krishna Khadka

Network Security Engineer | Internet Security Researcher | Enterprise Network Architect

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Education

University of Twente, The Netherlands	03/2023–Present (Expected 2026)
PhD Candidate , Computer Science. Researching Internet-scale network security with emphasis on routing security, DDoS resilience, and resilient Internet infrastructure.	
University of Crete, Greece	2016
MSc , Computer Science.	
Tribhuvan University, Nepal	2015
MSc , Computer Engineering.	
Tribhuvan University, Nepal	2022
Bachelor of Law (LLB) .	
Tribhuvan University, Nepal	2009
Bachelor of Computer Engineering .	

Research Experience

University of Twente — Doctoral Researcher	2023–Present
- Designed Internet-scale measurement methodologies for assessing the security of BGP and Autonomous Systems. - Designed and implemented novel methodologies to detect exposed BGP routers and characterize global DDoS scrubbing deployments. - Published and presented research at international networking and Internet measurement venues. - Presented research findings at international networking forums, including RIPE and IETF. - Assisted with the Advanced Networking course, with a focus on data-plane programming using P4.	

Publications

Detecting and Characterizing Exposed BGP Routers	
Shyam Krishna Khadka, Suzan Bayhan, Ralph Holz, Cristian Hesselman	TMA '26
Detecting and Characterizing DDoS Scrubbing from Global BGP Routing	
Shyam Krishna Khadka, Suzan Bayhan, Ralph Holz, Cristian Hesselman	PAM '26
Prefix Top Lists Reloaded: A Temporal Prefix Ranking Dataset	
Savvas Kastanakis, Rick Fontein, Shyam Krishna Khadka, Ebrima Jaw, Cristian Hesselman, Mattijs Jonker	PAM '26
Critical BGP Prefixes: A Measurement-based Investigation of Critical Infrastructure Security	
Savvas Kastanakis, Shyam Krishna Khadka, Ebrima Jaw, Cristian Hesselman	IOCRCI '26
A First Look at the Adoption of BGP-based DDoS Scrubbing Services	
Shyam Krishna Khadka, Suzan Bayhan, Ralph Holz, Saeedeh Shokoohi, Marinho Barcellos, Cristian Hesselman	CNSM '25
Assessing the Security of Internet Paths	
Shyam Krishna Khadka, Suzan Bayhan, Ralph Holz, Cristian Hesselman	ANRW '24
Towards Security Transparency of Autonomous Systems on the Internet	
Shyam Krishna Khadka, Ralph Holz, Cristian Hesselman	ESORICS '23

Professional Experience

Nepal Telecom — Senior Engineer	2010–2022
Progressed from Software Developer and Network/System Engineer to Senior Engineer at Nepal's largest telecommunications provider.	
- Led technical evaluation and deployment of SD-WAN, HCI, EDR, Wi-Fi 6, next-generation firewalls, and enterprise networking solutions involving Cisco, Juniper, Huawei, Check Point, Palo Alto, VMware, and Nutanix. - Managed routing, switching, and Linux system administration. - Handled technical procurement, compliance testing, and network operations. - Developed internal web applications and services using Java, ASP.NET, JavaScript, and Nginx.	
CloudFactory — Software Developer	2009–2010

- Developed web applications using Ruby on Rails and built database-driven applications.
- Worked in Agile software development teams and collaborated with international engineering teams.

Selected Talks & Outreach

- Presented research on BGP security and DDoS resilience at RIPE and IETF forums.
- Published research summaries through APNIC and RIPE Labs.
- Discussed research findings on DDoS scrubbing and Internet routing security in an industry podcast hosted by FastNetMon.

Awards & Recognition

- IRTF Diversity Grant, IETF 126, Vienna, 2026.
- One of the three best measurement blogs for 2025, APNIC.
- Erasmus Mundus Scholarship, University of Crete.
- Faculty Topper, Bachelor of Computer Engineering.
- Full Scholarship, Bachelor and MSc in Computer Engineering.

Technical Skills

Networking: BGP, OSPF, MPLS, routing, switching, SD-WAN, Wi-Fi 6, NGFW, CCNP training

Security: NGFW, XDR, firewalls, BGP security, DDoS mitigation

Programming: Python, Java, Ruby, C/C++, JavaScript

Systems: Linux, VMware, HCI, virtualization, Git, RHCE training

Research: Internet measurement, network security, data analysis, technical writing and presentations

Languages

English — Professional proficiency

Nepali — Native

Dutch — Basic (A2)